PERMANENCE OF SYNTOMIC/SMOOTH/\'ETALE ALONG A SURJECTIVE SYNTOMIC (RESP. SMOOTH, \'ETALE) F

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four Morphisms of Spaces lemmas and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- \'Etale locality of syntomic/smooth/\'etale morphisms. Stacks Project, Morphisms of Spaces, Lemmas spaces-morphisms-lemma-syntomic-local, spaces-morphisms-lemma-smooth-local, spaces-morphisms-lemma-etale-local. Justifies the reduction to affine B and Y.
- Openness of fppf morphisms on points. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-fppf-open. Lets the proof pick U with $U \to Y$ surjective.
- Scheme-side permanence. Stacks Project, Descent, Lemma descent-lemma-syntomic-smooth-etale-permanence. Concludes for the scheme diagram.

Source reference. Original reference 03ZA: sources/stacks-chow/tags.txt, source lines 5213–5213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

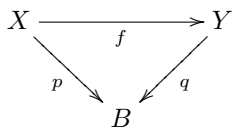
Source reference. Original reference 03ZF: sources/stacks-chow/tags.txt, source lines 5218–5218. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03XT: sources/stacks-chow/tags.txt, source lines 5160–5160. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 042S: sources/stacks-chow/tags.txt, source lines 5339–5339. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KM: sources/stacks-chow/tags.txt, source lines 3377–3377. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let*



be a commutative diagram of morphisms of algebraic spaces over S. Assume that

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(1) *f is surjective, and syntomic (resp. smooth, resp. étale),*
(2) *p is syntomic (resp. smooth, resp. étale).*

Then q is syntomic (resp. smooth, resp. étale).

Proof. We deduce this from the analogue for schemes. Namely, the problem is étale local on B and Y (Morphisms of Spaces, Lemmas 03ZA, 03ZF, and 03XT). Hence we may assume that B and Y are affine schemes. Since $|X| \to |Y|$ is open (Morphisms of Spaces, Lemma 042S), we can choose an affine scheme U and an étale morphism $U \to X$ such that the composition $U \to Y$ is surjective. In this case the result follows from Descent, Lemma 02KM. □